

Introduction!

CSE 1112: Structured Programming Language - Week 1

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Hello and Good Noon!



Hello and Good Noon!



A Little Introduction 😊

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- Lecturer (Feb 2019 – Jul 2023)
Department of Computer Science and Engineering
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For any **queries**, send an email to: raiyan@cse.uiu.ac.bd or visit my office (Room 636D) during counseling hours.

Weekly Schedule (counseling hour): [Click Here](#)





A Little Introduction, your turn 😊

Introduce yourself with:

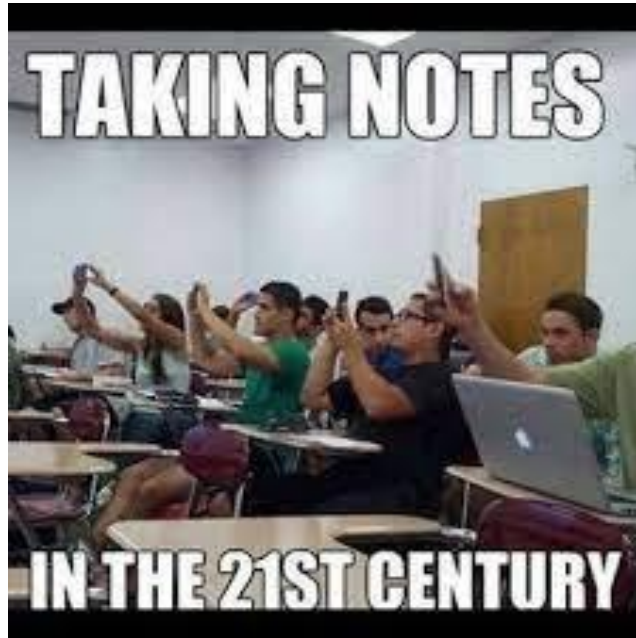
1. Your name
2. CGPA (no, kidding)
3. What you are liked the most about programming in ICS.
4. And, what you dislike the most.

Welcome to SPL!

Before we start, here's a (hopeful) promise:

If you follow my class closely, you will NOT need anything else for this course*.

*But



1. Pay attention and actually take notes, not pics.



2. Don't talk during the lectures.

*But

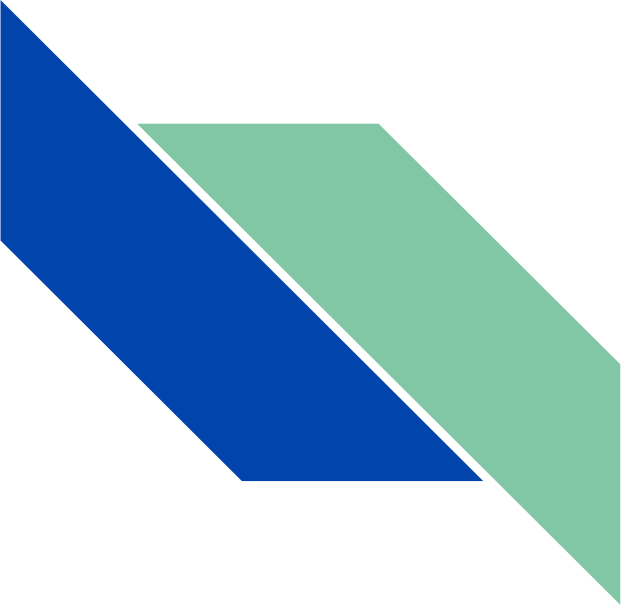


3. Bring a bed if you want to sleep. Promise, I'll let you.

*But

4. Be present in class. For absence due to valid reasons, send an email within the class day to raiyan@cse.uiu.ac.bd.

Now, on with the classes!



CSE 1112: SPL

Recap on Fundamentals of Problem Solving and Basic C Syntax



Lec Raiyan, Dept of CSE, UIU
Email: raiyen@cse.uiu.ac.bd

Week 1

What We know Computers to Be



What we (endlessly) browse the internet on!

What We know Computers to Be



What we game on!

What We know Computers to Be



What we (sometimes) work on!



What We know Computers to Be

So, computers are something we -

- Use for work
- Use for entertainment
- Use for information
- Use for business And whatnot!



What We know Computers to Be

But they are also the machines that-

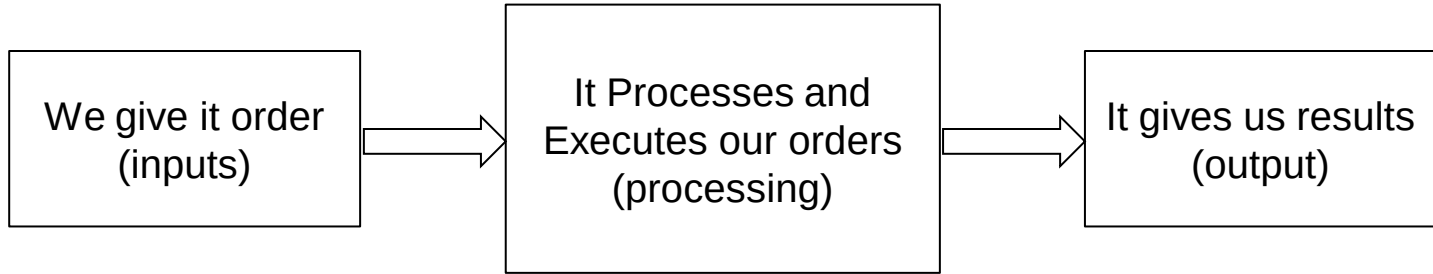
- Control the temperature we set on our AC.
- Are inside our calculators, microwave ovens and refrigerators
- Control complex production lines at factories.
- Land a rocket on the moon and back!



What are (Digital) Computers?

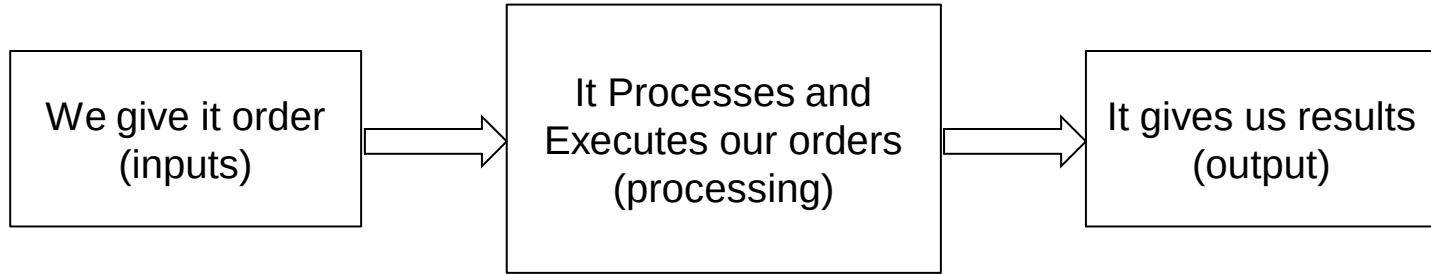
- A computer is a **machine** that can be instructed to carry out sequences of arithmetic or logical operations automatically via computer programming.
- An electronic **device** for storing and **processing** data, typically in binary form, **according to instructions given to it** in a variable program.
- A programmable **electronic device** designed to **accept data, process** prescribed mathematical and logical operations at high speed, and **display the results** of these operations.

So how do we define Computers?



Let's Try out the examples to fit this definition!

So how do we define Computers?

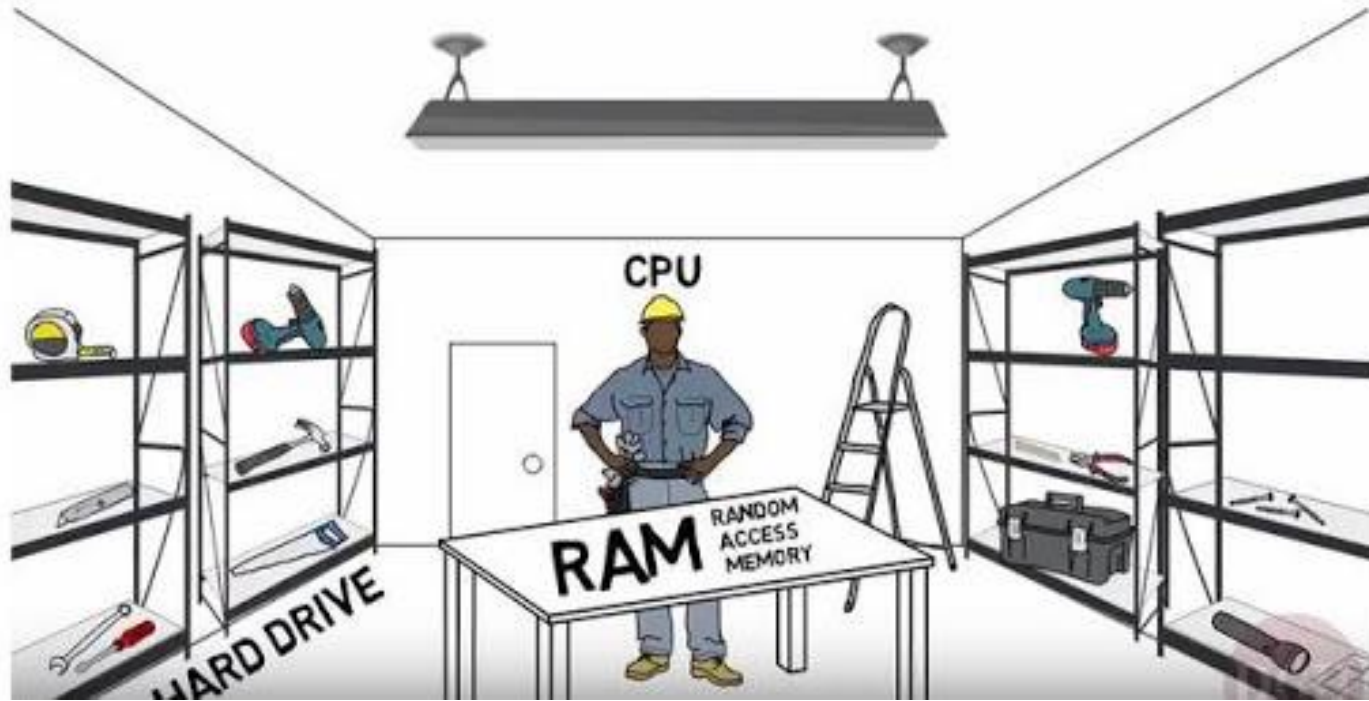


1. Keyboards
2. Mouse
3. Touchpads
4. Touch Screens
5. Gaming Controllers
6. Joystick
7. Scanners
8. Microphones (google voice assistant!)

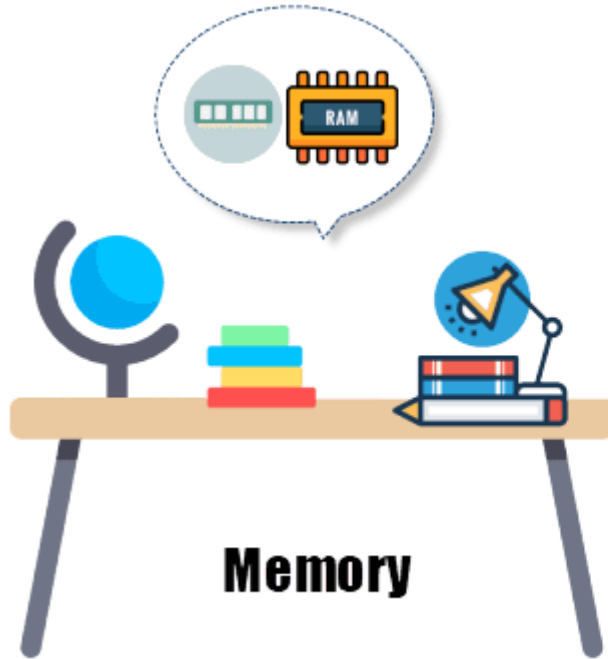
1. CPU
2. GPU
3. Memory (RAM/ROM)
4. Storage (SSD/HDD)
5. External Storage (USB, CD etc.)

1. Monitors
2. TVs
3. Screen
4. Small Displays
5. Printers
6. Speakers!

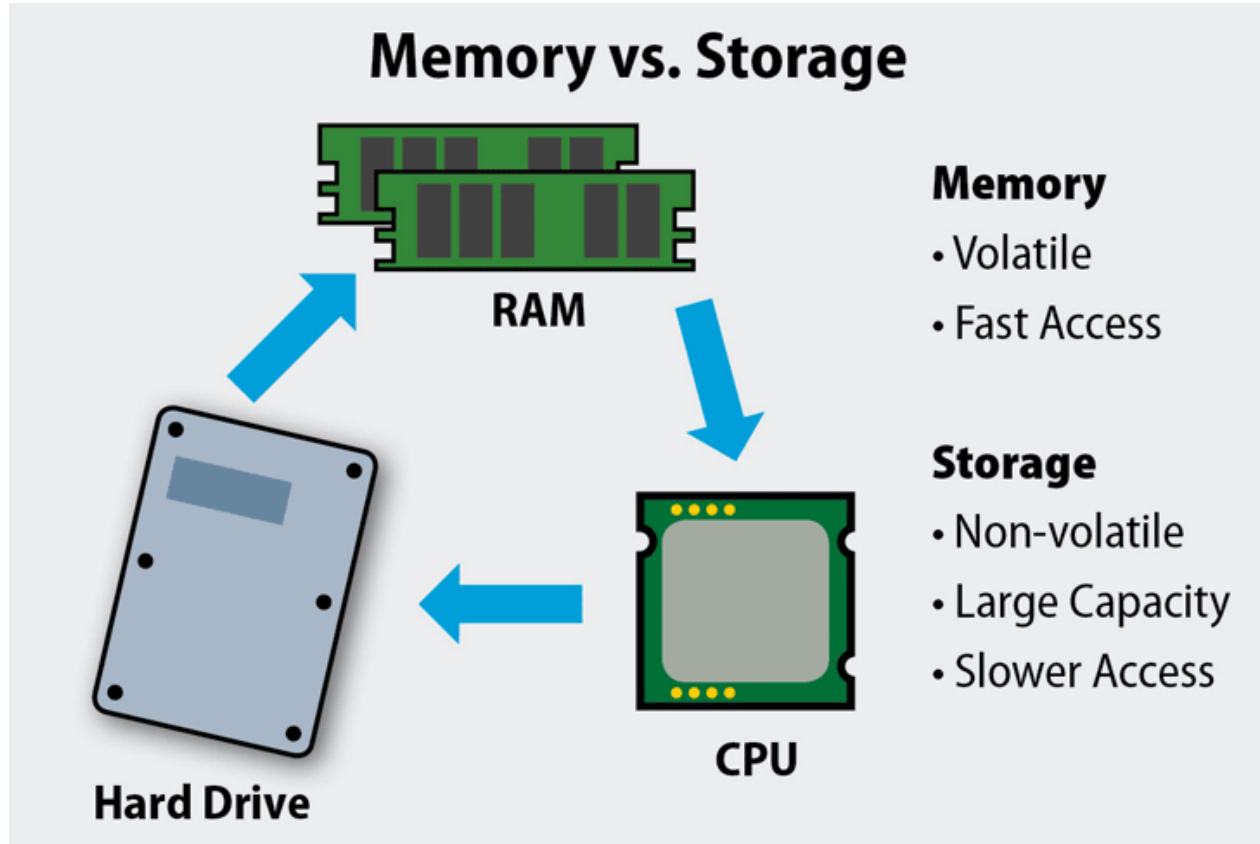
Computer Systems - A Metaphor



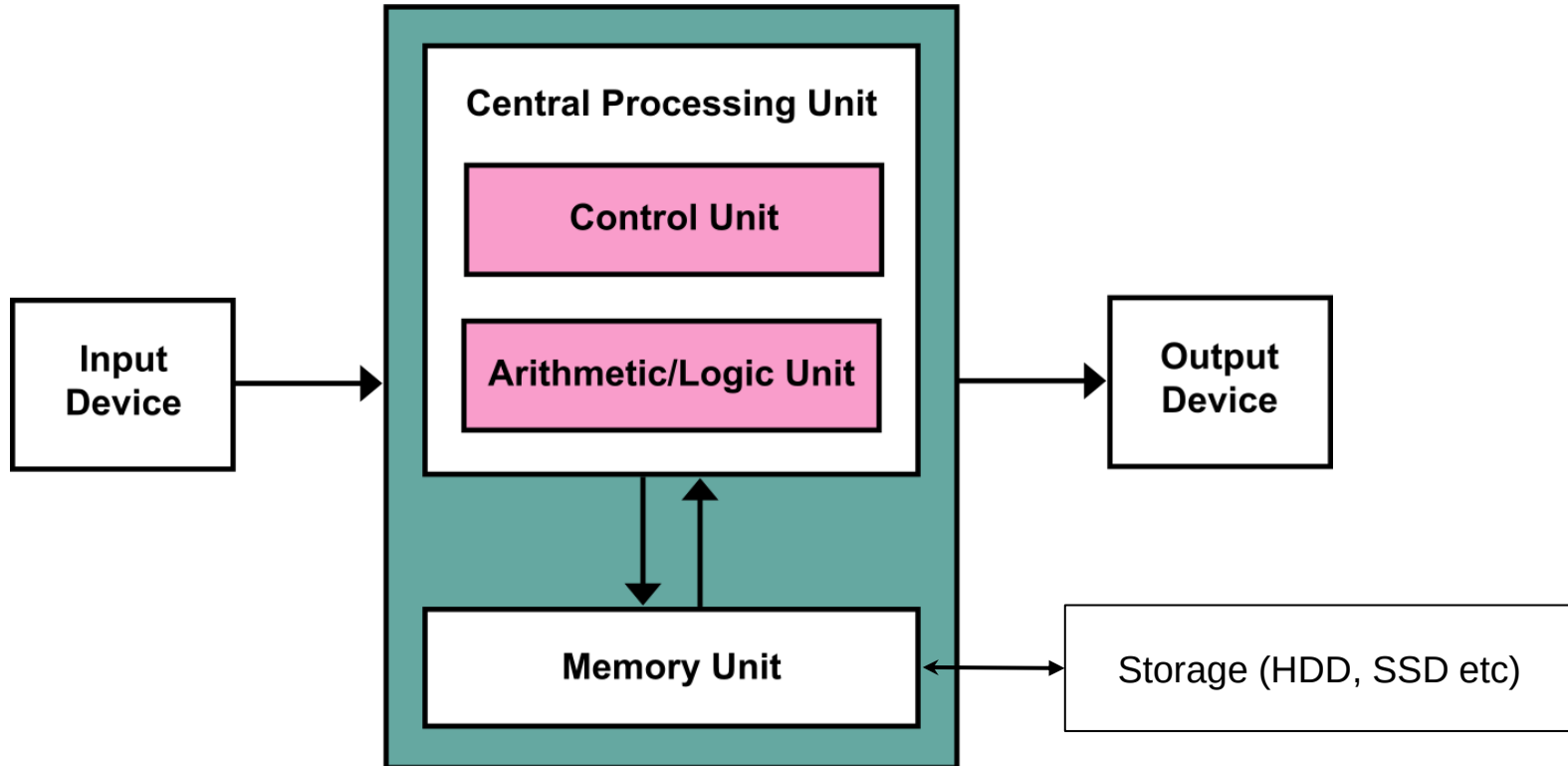
Computer Systems - Memory vs Storage



Computer Systems - Memory vs Storage



Computer Systems - Von Neumann Architecture





Some Simple Tasks

1. Do you know how to find out your PC's IP?
2. What is an IP address?
3. How do you actually remove a program from your PC?
4. Check which “devices” are installed on your PC.
5. What are device “drivers”?



So Computer Seems like a very smart machine that can
do a lot of Stuff!

But how does it do it?

It Processes and
Executes our orders
(processing)

But How



It doesn't know how to do anything!



It's really dumb!



Unless 'you' make it smart!



But how do you make it smart?

You tell it what to do!

And you do it with **computer programs!**

But again, computers are too dumb to understand you directly!

You gotta speak their language!

So you've got to speak in **computer programming!**



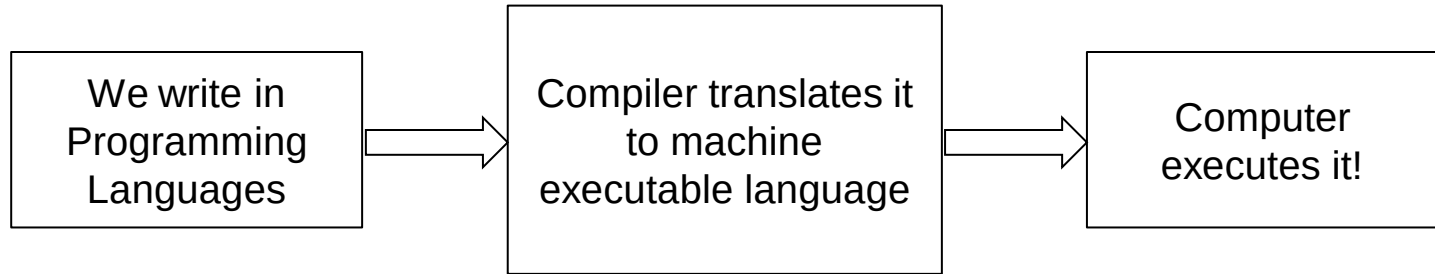
Computer Programs

Computer programming is the process that professionals use to write code that instructs how a computer, application or software program performs. At its most basic, computer programming is a set of instructions to facilitate specific actions.

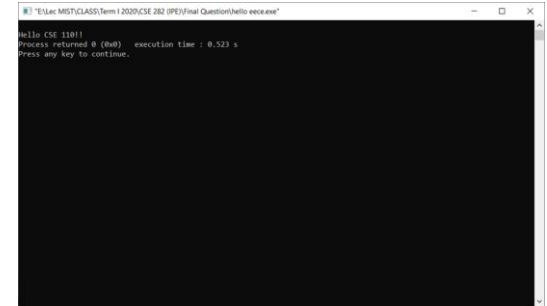
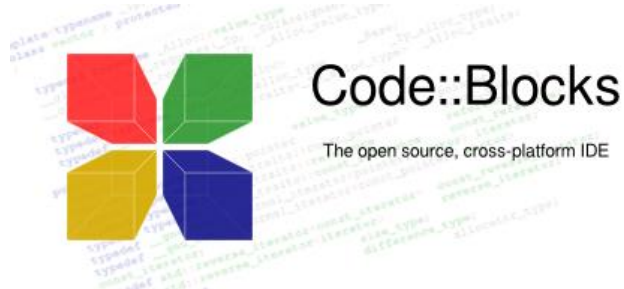
So, it -

- **Tells** the computer **how to** do/**solve a specific task**
- Gives Detailed **step by step instructions** to computers
- These instructions are **easily understandable** to computers.
- Just Like human languages, there are **many programming languages**, each better suited for a particular type of task.
- Written (mostly) **by humans**.

So how do we make a computer 'smart'?

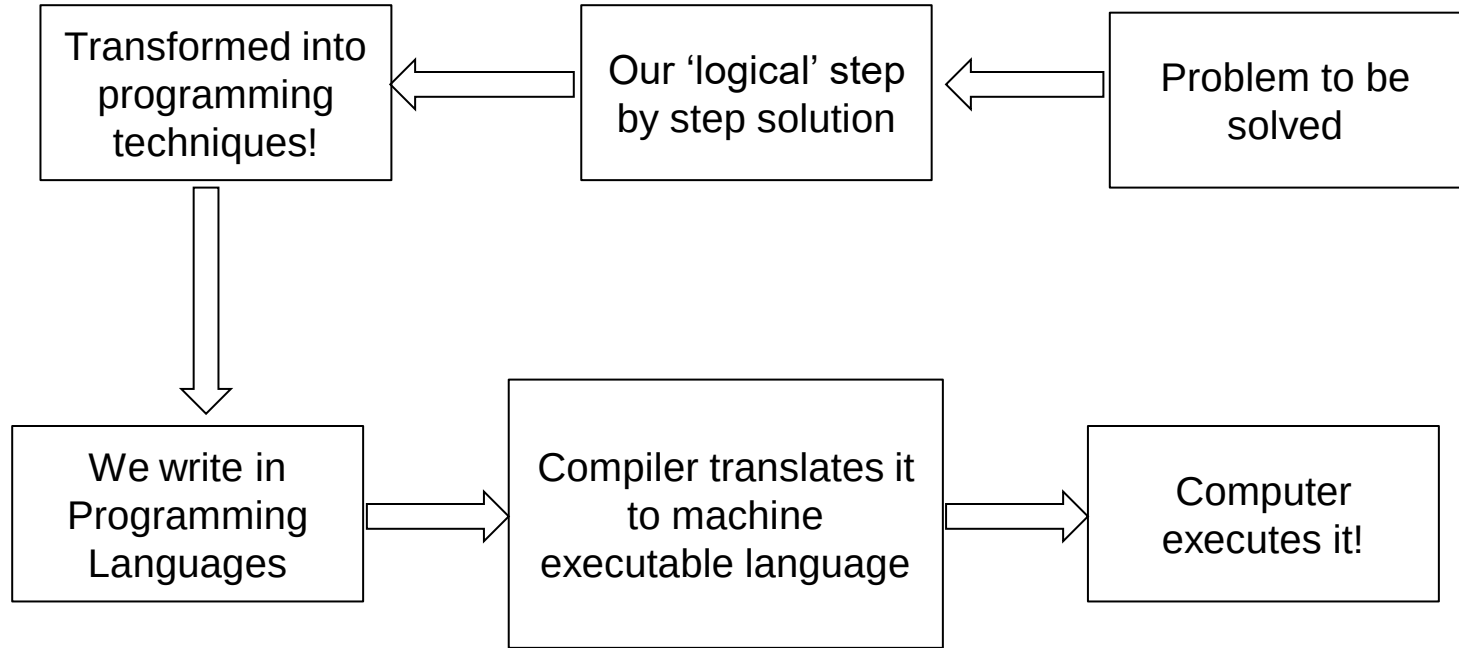


C/C++

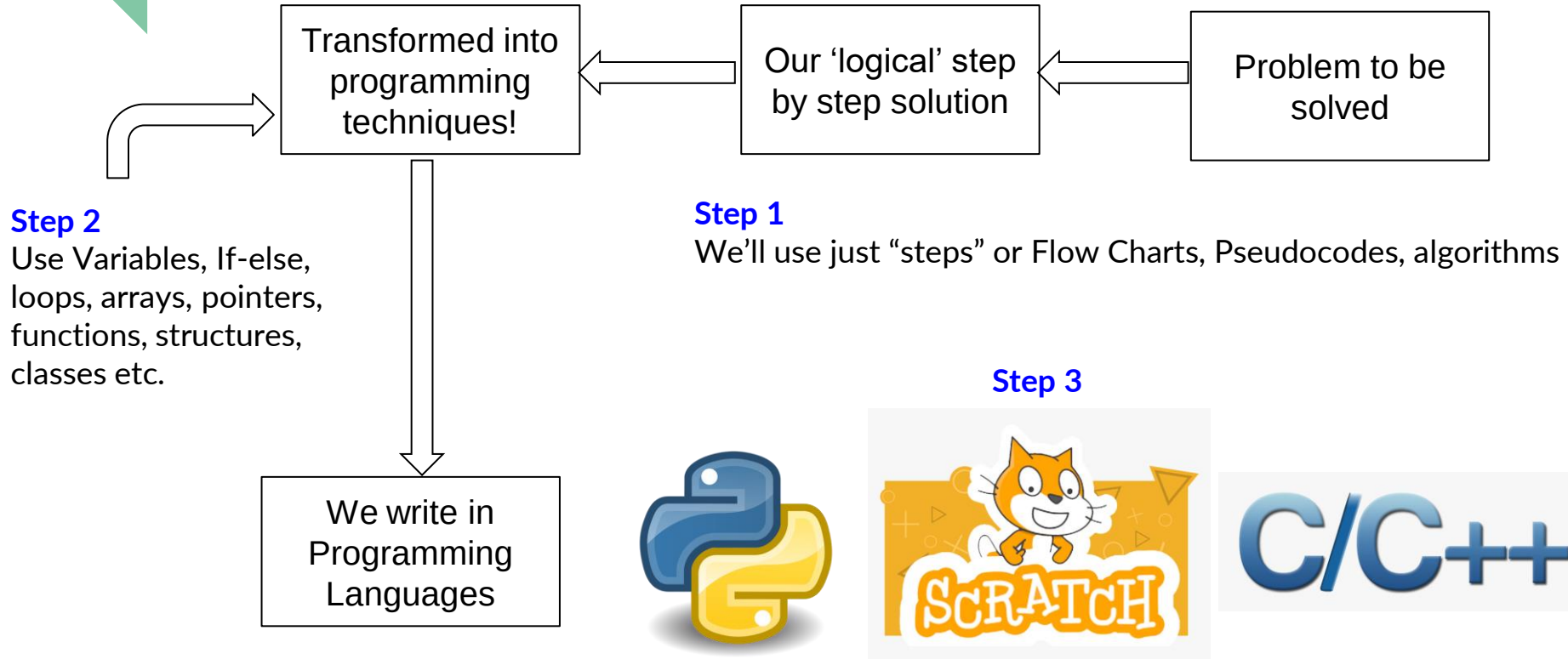


IDE + C/C++ compiler

So how do we make a computer 'smart'?



So how do we make a computer 'smart'?





What we hope to Achieve in CSE 115

To be able to create our own programs that can make the computer do a WIDE range of things!

A COVID Symptom Detection Tool: [Here](#)

A Library management System: [Here](#)

A Personal Website (maybe): TBD

You **should (hopefully)** be able to write such programs yourself after this course (CSE 115 + 116).



Let's do it!

- Download the compiler + IDE
- Know the basics of C programming
- Write our first program
- Solve our first problems using computers!
 - Finding a real life problem
 - Breaking it down to “Steps”
 - Transforming it to programming techniques
 - Coding it!



Let's do it!

- Download the compiler + IDE

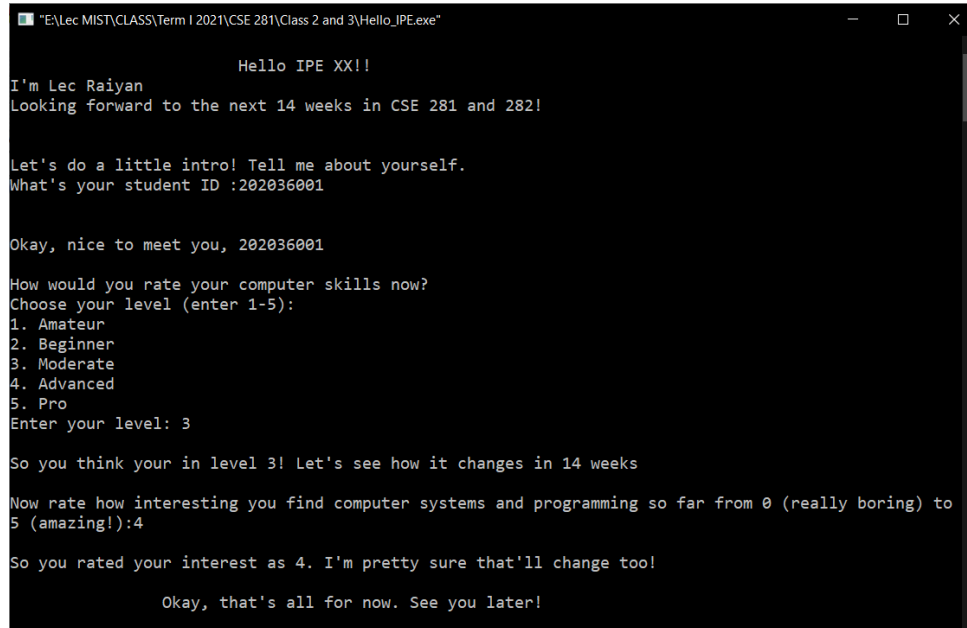
<http://www.codeblocks.org/downloads>

- For codeblocks installation guide, check:

<https://www.youtube.com/watch?v=GWJqsmitR2I&t=303s>

Example 1 - A Simple Conversation

Let's write a program (a chatbot) to have a simple conversation with the user. In time, we'll make more of a 'intelligent' chatbot!

A screenshot of a terminal window titled "E:\Lec MIST\CLASS\Term I 2021\CSE 281\Class 2 and 3\Hello_IPE.exe". The chatbot, "Lec Raiyan", initiates a conversation by saying "Hello IPE XX!!" and "Looking forward to the next 14 weeks in CSE 281 and 282!". It then asks for the user's student ID, which is "202036001". The chatbot responds with "Okay, nice to meet you, 202036001". It then asks the user to rate their computer skills from 1 to 5, with options: 1. Amateur, 2. Beginner, 3. Moderate, 4. Advanced, 5. Pro. The user enters "3". The chatbot says "So you think you're in level 3! Let's see how it changes in 14 weeks". It then asks the user to rate how interesting they find computer systems and programming from 0 (boring) to 5 (amazing), with the user entering "4". The chatbot responds "So you rated your interest as 4. I'm pretty sure that'll change too!". Finally, it says "Okay, that's all for now. See you later!".

```
"E:\Lec MIST\CLASS\Term I 2021\CSE 281\Class 2 and 3\Hello_IPE.exe"

Hello IPE XX!!

I'm Lec Raiyan
Looking forward to the next 14 weeks in CSE 281 and 282!

Let's do a little intro! Tell me about yourself.
What's your student ID :202036001

Okay, nice to meet you, 202036001

How would you rate your computer skills now?
Choose your level (enter 1-5):
1. Amateur
2. Beginner
3. Moderate
4. Advanced
5. Pro
Enter your level: 3

So you think you're in level 3! Let's see how it changes in 14 weeks

Now rate how interesting you find computer systems and programming so far from 0 (really boring) to
5 (amazing!):4

So you rated your interest as 4. I'm pretty sure that'll change too!

Okay, that's all for now. See you later!
```

[Check out the solution in Hello_IPE.c](#)



What we learnt

1. Writing a simple program in c.
2. What header files are (stdio.h in our program)
3. How to use printf().
4. How to use scanf().
5. Some basic idea on how to store an input and show it.
6. Making a little chatbot program in C!



Example 2 - Age Calculator

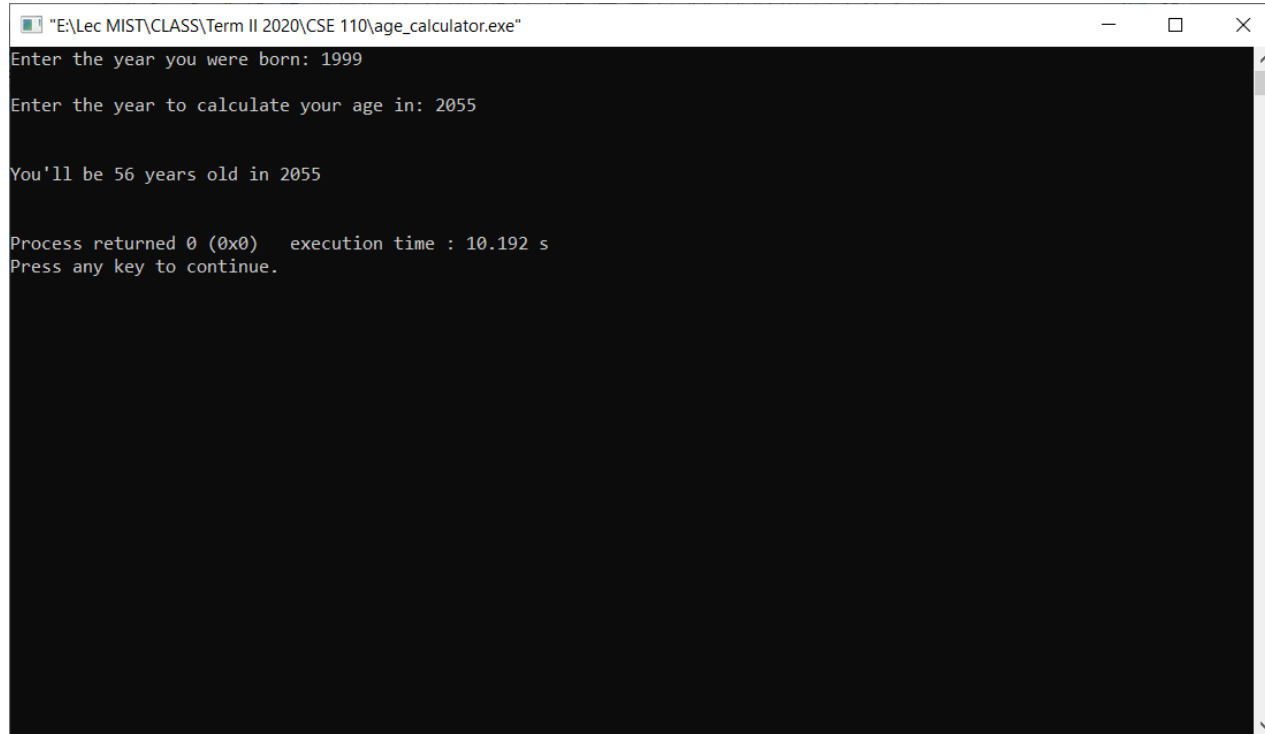
Make a Age Calculator that takes the birth year as input from the user and tells them how old they'll be in a particular year!

Input (To be taken from users)	Output
Enter The year you were born: 1999 Enter the year to calculate your age in: 2055	You'll be 56 years old in 2055.

[Check out the solution in age_calculator.c](#)

Example 2 - Age Calculator

Here's what the output should look like:



```
"E:\Lec MIST\CLASS\Term II 2020\CSE 110\age_calculator.exe"
Enter the year you were born: 1999
Enter the year to calculate your age in: 2055

You'll be 56 years old in 2055

Process returned 0 (0x0)   execution time : 10.192 s
Press any key to continue.
```



Example 2 - Age Calculator

Let's try out the steps for writing programs!

Step 1	Step 2
Ask the user when he/she was born	printf()
Store their answer (in birth_year)	scanf() + use an int variable
Ask and store the year he/she wants to calculate the age in (in calc_year)	printf() + scanf() + use an int variable
Do calc_year - birth_year to find age	Arithmetic operation (-)
Show them the result!	printf()

Example 3 - Simple Interest

Make a simple interest calculator for bank accounts that takes the principal, interest rate and time as input and finds the interest accordingly.

```
"E:\Lec MIST\CLASS\Term II 2020\CSE 110\simple_interest.exe"
Enter the principal you have in your account:45000
Enter your bank's interest rate:5
Enter the time this principal has been in the bank (years):5

Simple Interest Calculation for your account

Principal Amount = BDT 45000.00
Rate = 5.00
Time = 5 years
Simple interest = 11250.00

Process returned 27 (0x1B) execution time : 10.084 s
Press any key to continue.
```

[Check out the solution in simple_interest.c](#)



Example 3 - Simple Interest

Let's try out the steps for writing programs!

Step 1	Step 2
Ask and store the principal amount	printf() + scanf() + use an int variable
Ask and store the interest rate	printf() + scanf() + use another int variable
Ask and store the time	printf() + scanf() + use an int variable
Do $(\text{principal} * \text{rate} * \text{time}) / 100.0$ to find simple interest	Arithmetic operation + store in an int variable
Show the result!	printf()

Example 4 - Worker's Salary

Let's calculate a garment (or any) worker's salary calculator. Takes in daily rate and days worked, then calculates monthly salary. Then also takes in overtime hourly rate and hours worked to find salary with overtime.

```
"E:\Lec MIST\CLASS\Term II 2020\CSE 110\Codes\intro_salary.exe"

Salary Calculator
Developed in CSE 106 Sessional
MIST 2020

Enter your basic salary per day: 450
Enter how many days you worked last month: 26

Your basic salary for last month is: 11700.00

How many hours did you work overtime last month? : 24

Enter your overtime rate: 85

Your basic salary for last month (with overtime pay) is: 13740.00
Process returned 0 (0x0)   execution time : 30.558 s
Press any key to continue.
```

[Check out the solution in intro_salary.c](#)



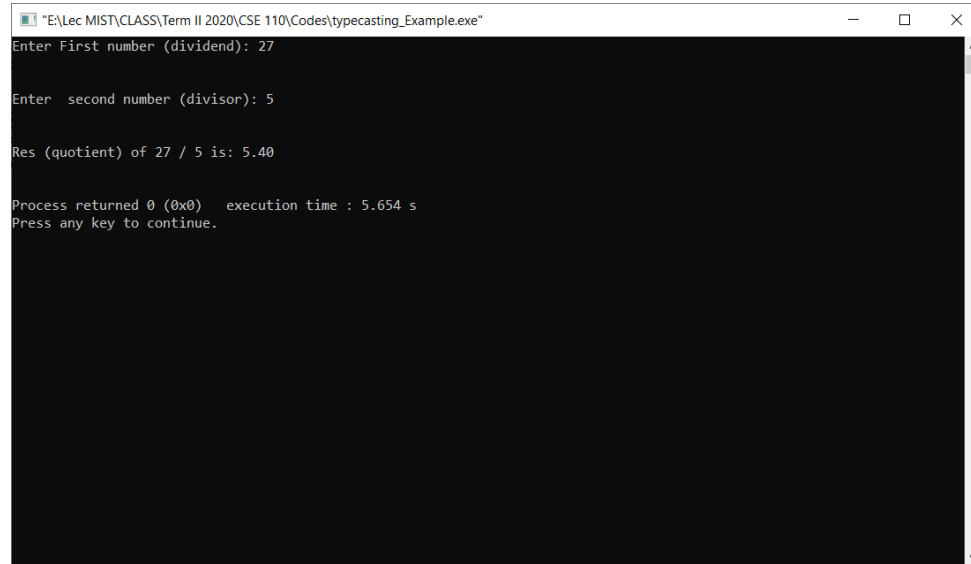
Example 4 - Worker's Salary

Let's try out the steps for writing programs!

Step 1	Step 2
Ask and store the basic daily salary	printf() + scanf() + use a float variable
Ask and store how many days the worker worked last month	printf() + scanf() + use an int variable
Calculate the Basic salary and show	Arithmetic operation + printf()
Ask and store overtime hours worked by the worker last month	printf() + scanf() + use an float variable
Ask and store overtime rate per hour	printf() + scanf() + store in a float variable
Calculate the OT salary (basic + overtime salary) and show.	Arithmetic operation + printf()

Example 5 - Typecasting

Let's try to divide one integer number by another (i.e $27/5$) and try to store the result in a float variable. Would we be alright without typecasting?



```
"E:\Lec MIST\CLASS\Term II 2020\CSE 110\Codes\typecasting_Example.exe"
Enter First number (dividend): 27

Enter second number (divisor): 5

Res (quotient) of 27 / 5 is: 5.40

Process returned 0 (0x0)   execution time : 5.654 s
Press any key to continue.
```

[Check out the solution in typecasting_example.c](#)



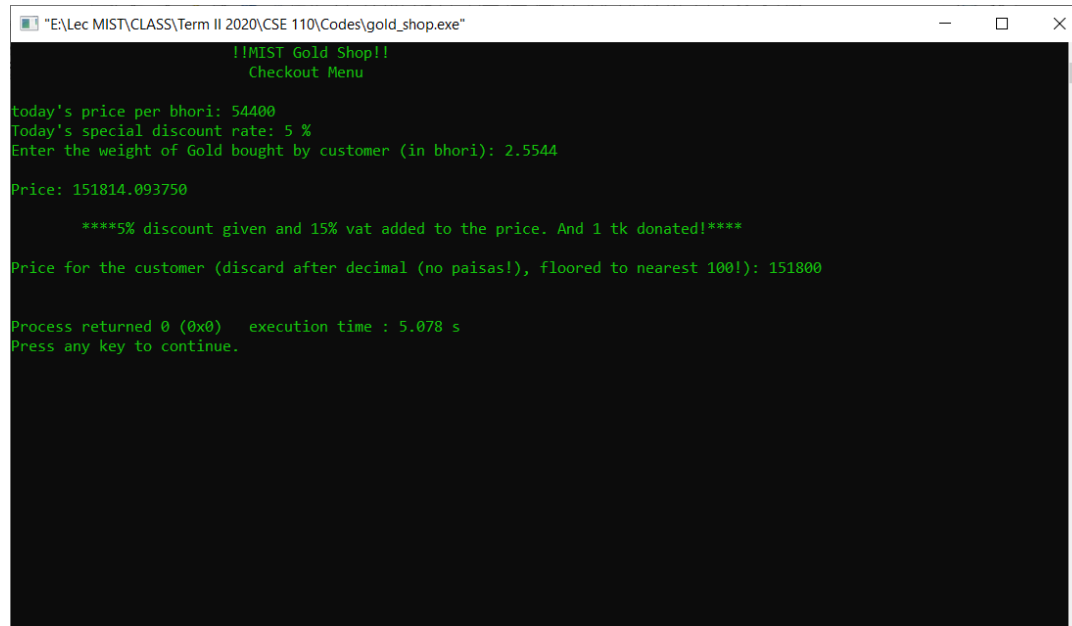
Example 5 - Typecasting

Let's try out the steps for writing programs!

Step 1	Step 2
Ask and store the num1(dividend)	printf() + scanf() + use a int variable
Ask and store the num 2 (divisor)	printf() + scanf() + use an int variable
Calculate the quotient	Arithmetic operation (a/b)
But we need to convert to float since the result of division of two integers may be decimal!	Have to Use typecasting!
Show the result of div (quotient)	printf()

Example 6 - A Gold Shop!

Let's make a program that calculates the price of a gold ornament to be purchased by the customer. It takes input of purchased amount of gold, today's rate/bhori and discount rate. Then, Vat (15%) and the entered discount to be given to the customer. Final price should also be "floored" to the nearest hundredth value. Also, one tk per purchase to be donated.



```
"E:\Lec MIST\CLASS\Term II 2020\CSE 110\Codes\gold_shop.exe"

!!MIST Gold Shop!!
Checkout Menu

today's price per bhorl: 54400
Today's special discount rate: 5 %
Enter the weight of Gold bought by customer (in bhorl): 2.5544

Price: 151814.093750

      ****5% discount given and 15% vat added to the price. And 1 tk donated!****

Price for the customer (discard after decimal (no paisas!), floored to nearest 100!): 151800

Process returned 0 (0x0)   execution time : 5.078 s
Press any key to continue.
```



Example 6 - A Gold Shop!

Step 1	Step 2
Take input of the daily gold rate (per bhor) and today's discount rate	printf() + scanf() + use float variables
Take input from shopkeeper how much gold bought (in bhor)	printf() + scanf() + use a float variable
Calculate the raw price	Arithmetic operation (gold_rate*weight_bought)
Give discount according to rate	Arithmetic operation + typecasting
Add 15% VAT, donate 1 tk	Arithmetic operation
Show the actual price	printf()
Calculate and show the price for the customer (discard all paisas, floor to nearest 100!)	Explicit typecasting + Arithmetic operation + printf()



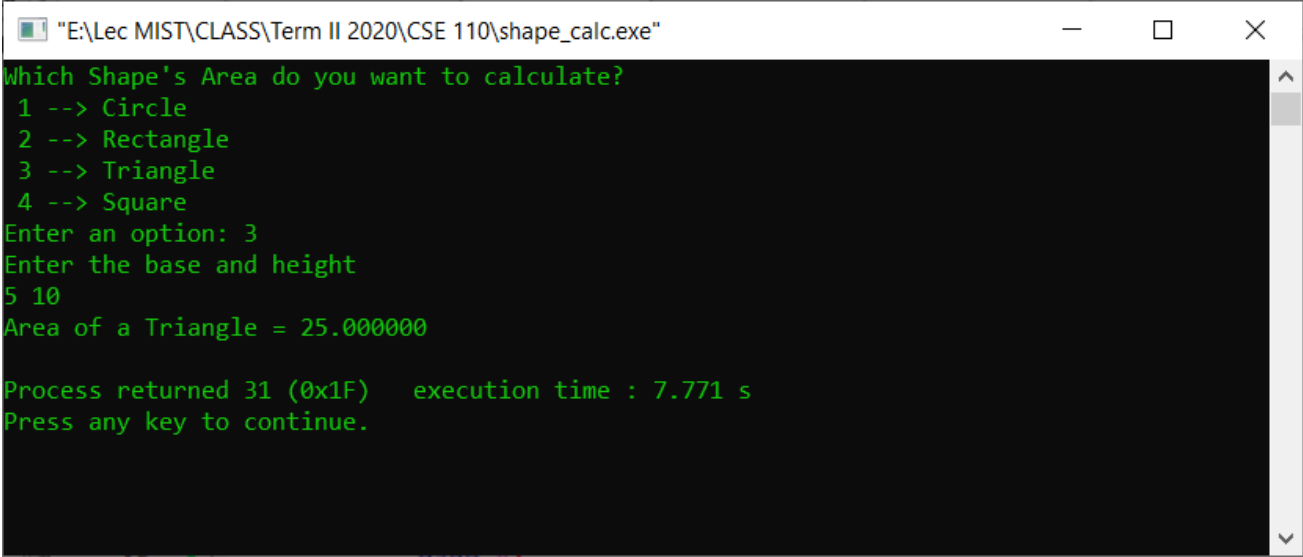
Escape Sequences for printf()

Escape Sequence	What it does
<code>\n</code>	Prints a new line. More specifically, moves the cursor to a new line.
<code>\t</code>	Moves the cursor one tab to the right.
<code>\r</code>	Moves the cursor to the beginning of the current line. Then whatever you write after <code>\r</code> will overwrite the previous characters. This is called carriage return.
<code>\b</code>	moves the cursor back one space. Then whatever you write overwrites what you had in that space. However, it can't go back to a previous line.
<code>\a</code>	Doesn't move the cursor. Just makes a “beep” or windows “alert” sound for that line.
<code>\</code>	Put <code>\</code> before any of the above escape sequences from above if you want to print the escape sequence itself rather than to move the cursor.
<code>%%</code>	However, if you want to print <code>%</code> itself just put another <code>%</code> before that. Putting <code>\</code> before <code>%</code> won't work.

Note: Escape characters take effect within that `printf()` (that line) only.

A Simple Task For Practice

Calculate the area of geometric shapes (circle, rectangle, square, triangle). Write separate programs for each shape's area calculation.



```
"E:\Lec MIST\CLASS\Term II 2020\CSE 110\shape_calc.exe"
Which Shape's Area do you want to calculate?
1 --> Circle
2 --> Rectangle
3 --> Triangle
4 --> Square
Enter an option: 3
Enter the base and height
5 10
Area of a Triangle = 25.000000

Process returned 31 (0x1F)   execution time : 7.771 s
Press any key to continue.
```

Do yourself for practice. Will solve in class next week

